

Day 17 — Mon 21 Sep

Countdown: 5 days to General English & GK

Method: start by recalling yesterday's topic with this page closed. Study the sections below, work every example by hand, then do 20 MCQs.

🕒 GK — General Science

Source: General/03_GK_CAPSULE.md

7.1 Physics

Measurement & units (SI):

Quantity	SI unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Force	newton	N
Energy / work	joule	J
Power	watt	W
Pressure	pascal	Pa
Frequency	hertz	Hz

Motion (Newton's laws):

1. **First law (inertia):** a body stays at rest or in uniform motion unless acted on by a force.

2. **Second law:** $F = ma$ (force = mass $\times$ acceleration).

3. **Third law:** every action has an equal and opposite reaction.

Equations of motion: $v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$.

Light: travels at $\sim 3 \times 10^8$ m/s; reflection & refraction; a **convex lens**

converges (used for magnifying, in the eye), a **concave lens** diverges (myopia correction); **VIBGYOR** is the spectrum; a prism disperses white light.

Sound: a **longitudinal mechanical wave**; needs a medium (cannot travel in vacuum); speed in air ~ 343 m/s $\triangle$ verify; audible range **20 Hz–20,000 Hz**; **ultrasound** is above 20 kHz.

Electricity: **Ohm's Law** $V = IR$; $P = VI$; series vs parallel circuits; a fuse protects against overload; AC vs DC.

7.2 Chemistry

Atomic structure: an atom = **protons (+)** and **neutrons** in the nucleus, with **electrons (–)** around it. **Atomic number** = number of protons; **mass number** =

protons + neutrons; **isotopes** have the same protons but different neutrons.

Acids, bases & pH:

Point	Detail
Acid	Releases H^+ ; sour; turns blue litmus red ; $pH < 7$
Base/Alkali	Releases OH^- ; bitter, soapy; turns red litmus blue ; $pH > 7$
Neutral	$pH = 7$ (pure water)
Neutralisation	acid + base $\rightarrow$ salt + water

Everyday chemistry: common salt = **NaCl**; baking soda = **NaHCO₃**; washing soda = **Na₂CO₃**; quicklime = **CaO**; the gas that turns lime-water milky = **CO₂**; rusting = iron + oxygen + water $\rightarrow$ hydrated iron oxide.

7.3 Biology

Human body systems (one line each): digestive, respiratory, circulatory (heart pumps blood; **RBCs carry O₂ via haemoglobin**), nervous (brain, spinal cord), skeletal, excretory (kidneys).

Nutrition & vitamins (classic MCQ block):

Vitamin	Deficiency disease	Source
A	Night blindness	Carrots, leafy veg
B1 (thiamine)	Beriberi	Whole grains
B12	Anaemia	Meat, dairy
C (ascorbic acid)	Scurvy	Citrus fruits, amla
D	Rickets (children) / osteomalacia	Sunlight, fish
K	Poor blood clotting	Leafy vegetables

Macronutrients: **carbohydrates** (energy), **proteins** (growth/repair), **fats** (energy store).

Diseases (cause type):

Disease	Caused by
Malaria	Protozoan (Plasmodium) — spread by female Anopheles mosquito
Dengue	Virus — Aedes mosquito
Tuberculosis	Bacterium (<i>Mycobacterium tuberculosis</i>)
Typhoid, Cholera	Bacteria
COVID-19, Influenza, AIDS	Viruses

Classification & ecology: living things $\rightarrow$ five kingdoms (Monera, Protista, Fungi, Plantae, Animalia); **photosynthesis** = plants convert CO_2 + water + sunlight $\rightarrow$ glucose + O_2 (chlorophyll); an **ecosystem** = producers $\rightarrow$ consumers $\rightarrow$ decomposers; a **food chain** transfers energy; biodiversity.

Science — likely exam questions

- [5] State Newton's three laws of motion.
- [5] Differentiate between an acid and a base with examples.
- [5] Name four vitamins and the disease caused by each deficiency.
- [5] What is photosynthesis? Write its word equation.

Science — MCQ traps (with answers)

Answers

1. SI unit of force? → **Newton (N)** 2. pH of a neutral solution? → **7** 3. Deficiency of Vitamin C causes? → **Scurvy** 4. Malaria is spread by? → **female Anopheles mosquito** (Plasmodium) 5. Audible frequency range for humans? → **20 Hz to 20,000 Hz** 6. Gas that turns lime-water milky? → **CO₂** 7. Ohm's law? → **V = IR** 8. Atomic number = number of? → **protons**

SECTION 8 — MENTAL ABILITY (with shortcuts + solved practice)

Intuition for the whole section. These are speed questions. The examiner is testing whether you can apply a *method* fast — so learn the shortcut, then drill. Every topic below gives you the **shortcut**, a **worked example**, and **practice with answers**.

● Core — routing, DNS, crypto

Source: *_Shared/CORE_05_ComputerNetworks.md*

10. Routing

Concept

Routing is deciding *which path* a packet should take; **forwarding** is the act of moving it out of the right interface. A router keeps a **routing table** whose entries pair a **destination network** with a **next hop**, plus a **metric** (the cost) and often an administrative distance. When a packet arrives, the router finds the **longest matching prefix** in the table and forwards accordingly; if nothing matches, it uses the **default route** (**0.0.0.0/0**), and if there is none, it drops the packet and returns an ICMP "destination unreachable".

	Static routing	Dynamic routing
Configured by	Administrator, manually	Learned automatically by a routing protocol
Adapts to failure	No — needs manual change	Yes — reconverges around a failure
Overhead	None (no protocol traffic, no CPU)	Bandwidth and CPU consumed by updates
Security	More secure/predictable	Updates can be forged unless authenticated
Suits	Small, stable networks: stub networks	Large, changing networks

Distance Vector vs Link State — the central comparison

Distance vector ("routing by rumour"): each router knows only the **distance** (metric) and **direction** (next hop) to each destination, and it learns them by **periodically sending its entire routing table to its directly connected neighbours**. It never sees the whole topology —

it simply trusts what its neighbours tell it. Algorithm: **Bellman–Ford**.

Link state: every router **floods** a small **Link State Advertisement** describing only **its own directly attached links** to **every router in the area**. Each router therefore builds an **identical complete map** of the topology (the link-state database), and then independently runs **Dijkstra's shortest-path-first algorithm** on that map to compute its own best paths.

Criterion	Distance Vector	Link State
Algorithm	Bellman–Ford	Dijkstra (SPF)
What is shared	The entire routing table	Only link states (LSAs) about own links
Shared with	Directly connected neighbours only	All routers in the area (flooding)
When shared	Periodically (RIP: every 30 s)	Only on change (plus periodic refresh)
Knowledge of topology	Only what neighbours report — no map	Complete map of the area
Convergence	Slow	Fast
CPU / memory	Low	High (must store the whole database and run SPF)
Bandwidth	High periodically (whole table)	Low steady-state (small triggered updates)
Loop problems	Count-to-infinity , routing loops	Essentially loop-free
Loop remedies	Split horizon, poison reverse, hold-down timers, triggered updates, max hop count	Not required
Hierarchy	Flat	Areas (OSPF area 0 = backbone)
Examples	RIP, IGRP	OSPF, IS-IS

The count-to-infinity problem — a favourite 5-marker. If router A's link to network X fails, A hears from neighbour B "I can reach X in 2 hops" (B learnt it from A in the first place). A now believes X is 3 hops away via B; B then hears 3 and says 4, and the metric "counts to infinity" while packets loop. Solutions:

- **Maximum metric** — RIP defines **16 = infinity/unreachable**, capping the count.
- **Split horizon** — never advertise a route back out of the interface you learnt it on.
- **Poison reverse** — actively advertise it back with metric 16 (unreachable).
- **Hold-down timers** — ignore new information about a downed route for a period.
- **Triggered updates** — send changes immediately rather than waiting for the timer.

Routing protocols to know

Protocol	Type	IGP/EGP	Metric	Max hops	Updates	Notes
RIP v1	Distance vector	IGP	Hop count	15 (16 = ∞)	Broadcast every 30 s , whole table	Classful — no VLSM/CIDR support

RIP v2	Distance vector	IGP	Hop count	15	Multicast 224.0.0.9, every 30 s	Classless — carries the mask; supports VLSM; authentication
IGRP / EIGRP	DV / advanced (hybrid)	IGP	Composite: bandwidth, delay, load, reliability	255	EIGRP: triggered, uses DUAL algorithm	Cisco proprietary (EIGRP now open)
OSPF	Link state	IGP	Cost = reference bandwidth / interface bandwidth	Unlimited	Triggered LSAs, multicast 224.0.0.5/6	Open standard , hierarchical areas , area 0 = backbone , fast convergence, VLSM
IS-IS	Link state	IGP	Cost	Unlimited	Triggered	Used in large ISP cores
BGP	Path vector	EGP	AS-PATH and policy attributes	—	Incremental, over TCP port 179	The routing protocol of the Internet ; routes between Autonomous Systems ; chosen for <i>policy</i> , not shortest path

IGP vs EGP: an **Interior Gateway Protocol** (RIP, OSPF, EIGRP, IS-IS) routes **within** a single **Autonomous System**; an **Exterior Gateway Protocol** (**BGP**) routes **between** autonomous systems. An **AS** is a set of networks under one administrative authority with a common routing policy, identified by an **AS number**.

Other routing concepts named in your syllabus

- **Flooding:** send every incoming packet out of every interface except the one it arrived on. Guarantees delivery and always finds the shortest path, but generates enormous duplicate traffic; controlled by a hop counter or sequence numbers. Used in LSA distribution and military/robust networks.
- **Shortest path routing (Dijkstra):** build a graph, greedily grow a set of nodes with known shortest distance from the source, relaxing edges as you go.
- **Hierarchical routing:** divide the network into regions; routers know full detail of their own region and only summaries of others. Cuts table size dramatically at a small cost in path optimality — the reason the Internet is routable at all.
- **Broadcast, multicast and anycast routing; spanning tree** for loop-free broadcast.
- **Virtual circuits vs datagrams** — see §12.
- **Concatenated virtual circuits** (explicitly in your Forensic syllabus): when a connection crosses several different networks, a virtual circuit is set up in each network and the circuits are **spliced/concatenated end to end** at the gateways, so the whole path behaves like one connection-oriented circuit. Contrast with the **connectionless internetworking** approach (what IP actually does), where each datagram is routed independently and may take different routes and arrive out of order. Concatenated VCs preserve ordering and allow resource reservation, but fail entirely if any constituent network fails and cannot route around failures.

Likely exam questions

- **[5]** Differentiate between static and dynamic routing.
- **[5]** Explain the count-to-infinity problem and any two solutions.
- **[5]** What is a routing table? What information does it contain?
- **[5]** Differentiate between RIP and OSPF.
- **[10]** Compare distance vector and link state routing algorithms in detail, naming the

underlying algorithm, what is exchanged, with whom, how often, and the convergence and loop characteristics of each.

- **[15]** Explain routing in computer networks: routing tables, static vs dynamic, distance vector (Bellman–Ford) and link state (Dijkstra) with an illustrative topology, the count-to-infinity problem and its remedies, and the roles of RIP, OSPF and BGP.

MCQ traps

Trap	Correct answer
"RIP maximum hop count?"	15 (16 means unreachable).
"OSPF metric?"	Cost (bandwidth-based), not hop count.
"Which algorithm does OSPF use?"	Dijkstra / SPF . RIP uses Bellman–Ford .
"Which is the Internet's routing protocol?"	BGP — a path vector EGP over TCP 179 .
"OSPF backbone area?"	Area 0 .
"RIPv1 supports VLSM?"	No — it is classful. RIPv2 does.
"Split horizon prevents?"	Routing loops / count-to-infinity.
"Distance vector sends what, to whom?"	Whole table , to neighbours only .

15. DNS — Domain Name System

Concept

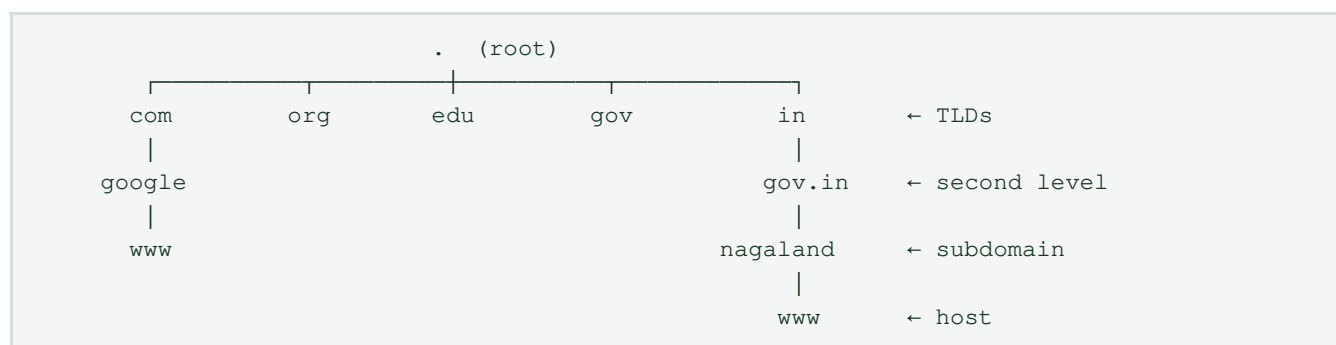
People remember `www.nagaland.gov.in`; routers only understand `104.18.32.7`. **DNS is the distributed database that translates between them.** Two design decisions define it:

1. **It is hierarchical**, so the name space can be delegated. Nobody has to know every name.
2. **It is distributed**, so no single server holds everything and no single failure is fatal.

The name space

An inverted tree again. Read a domain name **right to left**, from most general to most specific. A fully qualified domain name (FQDN) technically ends with a dot for the root:

`www.nagaland.gov.in.`



TLD type	Examples
gTLD (generic)	<code>.com</code> , <code>.org</code> , <code>.net</code> , <code>.edu</code> , <code>.gov</code> , <code>.mil</code> , <code>.int</code> , <code>.info</code> , <code>.biz</code>
ccTLD (country code)	<code>.in</code> , <code>.uk</code> , <code>.us</code> , <code>.jp</code> , <code>.cn</code>

Second level under ccTLD

.co.in, .gov.in, .ac.in, .nic.in

A **zone** is the portion of the name space an individual server is **authoritative** for — the unit of administrative delegation. A **domain** is a subtree of the name space. They coincide unless the domain delegates sub-zones.

Types of name server

Type	Role
Root name server	Top of the hierarchy; knows the authoritative servers for every TLD. There are 13 logical root server addresses (A–M), served by hundreds of physical machines via anycast
TLD server	Authoritative for a top-level domain; knows the servers for each second-level domain
Authoritative server	Holds the actual zone file with the real records for a domain. Primary/master holds the editable copy; secondary/slave copies it via a zone transfer (AXFR/IXFR, over TCP) for redundancy
Recursive resolver / local DNS server	The server your machine is configured to ask (usually your ISP's or 8.8.8.8). It does the legwork on your behalf and caches results
Caching-only server	Holds no zone of its own; just resolves and caches

Resolution — the process to describe

Recursive query: the client asks its local resolver and demands a *final answer* — "either the address or an error, but do not send me elsewhere." The resolver takes on all the work.

Iterative query: the resolver asks a server, which replies "I don't know, but ask *this* server" — a **referral**. The resolver then asks the next one, and so on.

Worked resolution of `www.example.com` (write this out step by step for 10 marks):

1. The browser checks its own cache, then the OS cache, then the local **hosts file** (`/etc/hosts`, or `C:\Windows\System32\drivers\etc\hosts`). Hit → done.
2. Miss → the host sends a **recursive** query to its configured **local DNS resolver**.
3. The resolver checks **its** cache. Miss → it queries a **root** server **iteratively**.
4. The root replies with a referral to the **.com TLD** servers.
5. The resolver queries a **.com** server → referral to the **authoritative** servers for `example.com`.
6. The resolver queries the authoritative server → it returns the **A record**:
`www.example.com → 93.184.216.34`.
7. The resolver **caches** the answer for its **TTL** and returns it to the client.

DNS uses **UDP port 53** for queries (small, fast, retried if lost) and **TCP port 53** for **zone transfers** and responses larger than 512 bytes.

Resource records — explicitly named in your Forensic syllabus

A resource record is a five-tuple: **Domain name, TTL, Class (IN), Type, Value**.

Type	Name	Purpose	Example
A	Address	Maps a name to an IPv4 address	<code>www.example.com. 3600 IN A 93.184.216.34</code>
AAAA	Quad-A	Maps a name to an IPv6 address	<code>IN AAAA 2606:2800:220:1::1</code>

NS	Name Server	Names the authoritative servers for the zone	<code>example.com. IN NS ns1.example.com.</code>
MX	Mail Exchange	The mail server for the domain, with a preference number (lower = higher priority)	<code>example.com. IN MX 10 mail.example.com.</code>
CNAME	Canonical Name	An alias pointing to another name	<code>ftp.example.com. IN CNAME www.example.com.</code>
PTR	Pointer	Reverse lookup — IP → name, in the <code>in-addr.arpa</code> domain	<code>34.216.184.93.in-addr.arpa. IN PTR www.example.com.</code>
SOA	Start of Authority	Zone parameters: primary server, admin email, serial number , refresh, retry, expire, minimum TTL. Exactly one per zone, and it must be first	
TXT	Text	Arbitrary text — used for SPF, DKIM, DMARC (email anti-spoofing) and domain verification	
SRV	Service	Locates the host/port for a named service	<code>_sip._tcp.example.com.</code>
HINFO	Host Info	CPU and OS description (rarely used; an information-disclosure risk)	
CAA	Cert Authority Auth.	Which CAs may issue certificates for the domain	

For the Forensic paper, note the investigative tools: `nslookup`, `dig`, `host`, `whois` (registrant details, registration and expiry dates, name servers), and reverse DNS via PTR. TXT records holding **SPF** and **DKIM** are what you check when analysing whether an email was spoofed.

Likely exam questions

- **[5]** What is DNS? Why is it needed?
- **[5]** Explain the hierarchical structure of the DNS name space with a diagram.
- **[5]** Differentiate between recursive and iterative DNS queries.
- **[5]** What are resource records? Explain A, MX, CNAME, NS and PTR.
- **[10]** Explain the Domain Name System: the name space, zones and domains, types of name server, resource records, and the complete resolution of `www.example.com` step by step.
- **[15]** Discuss DNS in detail — architecture, name servers, resource records, caching and TTL, recursive vs iterative resolution — and explain DNS-based attacks (cache poisoning, spoofing, tunnelling) and their countermeasures.

MCQ traps

Trap	Correct answer
"DNS port and transport?"	53 , UDP for queries, TCP for zone transfers.
"Record for IPv6?"	AAAA (A is IPv4).
"Record for a mail server?"	MX . For an alias → CNAME .
"Reverse lookup record?"	PTR , under <code>in-addr.arpa</code> .
"How many SOA records per zone?"	Exactly one .

"Lower MX preference means?"	Higher priority.
"Number of root server addresses?"	13 (A–M).
"Which query type does the client send to its resolver?"	Recursive .

18. Network security, cryptography and firewalls

Concept

Security is usually framed as the **CIA triad** plus two:

Goal	Meaning	Threatened by
Confidentiality	Only authorised parties can read the data	Eavesdropping, sniffing, traffic analysis
Integrity	Data is not altered in transit or storage	Modification, replay, MITM
Availability	Services are usable when needed	DoS / DDoS , ransomware
Authentication	Parties are who they claim to be	Spoofing, masquerade
Non-repudiation	A sender cannot later deny having sent	— (provided by digital signatures)

Attack classification — a reliable 5-marker:

	Passive attacks	Active attacks
Nature	Observe only; do not alter data	Alter data or disrupt service
Detectability	Very hard to detect (nothing changes)	Detectable
Prevention	Prevention is the goal (encryption)	Detection and recovery is the goal
Examples	Eavesdropping/sniffing , traffic analysis	Masquerade/spoofing , replay , modification , denial of service , MITM

Named threats to define in one line each:

Threat	Definition
Packet sniffing	Capturing traffic off the wire/air with a NIC in promiscuous mode (Wireshark, tcpdump). Trivial on a hub or Wi-Fi; needs ARP spoofing or port mirroring on a switch
Spoofing	Forging a source identity — IP spoofing , MAC spoofing , ARP spoofing , DNS spoofing/cache poisoning , email spoofing
Man-in-the-middle	Attacker relays and possibly alters traffic between two parties who believe they talk directly
Replay	Capturing a valid message and re-sending it later. Countered by nonces , timestamps , sequence numbers
DoS / DDoS	Exhausting a resource so legitimate users are denied service; distributed when launched from a botnet . Examples: SYN flood, ping of death, smurf, amplification
Phishing	Fraudulent messages inducing the victim to reveal credentials; spear phishing is targeted

Malware	Virus (attaches to a host file, needs a host and user action), worm (self-replicating, self-propagating over a network , needs no host file), Trojan horse (disguised as legitimate, does not self-replicate), ransomware , spyware , keylogger , rootkit , botnet , logic bomb
SQL injection	Injecting SQL through an input field (see the DBMS file)
XSS	Injecting script into a web page viewed by others

Virus vs worm is asked constantly: a **virus requires a host file and human action to spread**; a **worm is standalone and spreads by itself across a network**.

Cryptography

Encryption turns **plaintext** into **ciphertext** using an **algorithm (cipher)** and a **key**; **decryption** reverses it. **Kerckhoffs's principle**: the security must rest entirely in the **key**, never in the secrecy of the algorithm.

Symmetric (secret key / private key / conventional)

One key is used for both encryption and decryption, and both parties must have it.

- Fast — suitable for **bulk data**.
- **Two problems**: (1) **key distribution** — how do you get the key to the other party securely in the first place? (2) **key explosion** — $n(n-1)/2$ **keys** are needed for n parties to communicate pairwise. For $n = 100$ that is **4,950** keys.
- **Cannot provide non-repudiation** — since both parties hold the same key, either could have produced any message.
- Algorithms: **DES** (56-bit key, 64-bit block — **broken by brute force**), **3DES/Triple DES** (168/112-bit effective), **AES** (Rijndael; **128/192/256-bit keys**, 128-bit block — the current standard), **Blowfish**, **Twofish**, **IDEA**, **RC4** (stream, deprecated), **RC5**.
- Types: **block ciphers** (fixed-size blocks, modes ECB/CBC/CFB/OFB/CTR) and **stream ciphers** (bit/byte at a time).

Asymmetric (public key)

A mathematically related key pair: a **public key**, published to the world, and a **private key**, never disclosed. What one key encrypts, only the other can decrypt. Introduced by **Diffie and Hellman (1976)**.

The two modes — and getting these the right way round is the whole topic:

Goal	Encrypt with	Decrypt with	Why
Confidentiality	The receiver's PUBLIC key	The receiver's private key	Only the receiver holds the private key, so only the receiver can read it
Authentication / digital signature	The sender's PRIVATE key	The sender's public key	Only the sender could have produced it, so it proves origin and gives non-repudiation

Algorithms: **RSA** (Rivest–Shamir–Adleman, 1977; security rests on the difficulty of **factoring large primes**), **Diffie–Hellman** (key *exchange* only, not encryption; based on the **discrete logarithm** problem), **ECC** (elliptic curve — equivalent strength with much shorter keys), **DSA/ElGamal**.

Comparison table:

Criterion	Symmetric	Asymmetric
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Digital signature vs encryption:

	Digital signature	Encryption
Provides	Authentication, integrity, non-repudiation	Confidentiality
Key used to create	Sender's private key	Receiver's public key
Key used to verify/open	Sender's public key	Receiver's private key
Is the message hidden?	No — signing does not conceal	Yes

Digital certificate and PKI: a public key alone proves nothing — how do you know it belongs to your bank and not to an attacker? A **Certificate Authority (CA)** issues a **digital certificate (X.509)** binding an identity to a public key and **signs it with the CA's own private key**. Your browser trusts a set of root CAs, so it can verify the chain. That whole apparatus — CAs, certificates, registration authorities, revocation lists (CRL/OCSP) — is the **Public Key Infrastructure (PKI)**.

Firewalls

A **firewall** is a hardware or software barrier placed between a trusted internal network and an untrusted external one, which **inspects traffic and permits or denies it according to a rule set (ACL)**. It enforces the network's security policy at a single choke point.

Type	Layer	How it works	Strengths	Weaknesses
Packet filter	3–4	Examines each packet's source/destination IP, port, protocol and flags against static rules; stateless	Fast , cheap, transparent	No context — cannot tell a reply from an unsolicited packet; cannot see payloads; vulnerable to spoofing and fragmentation attacks
Stateful inspection	3–4	Maintains a state table of active connections ; permits inbound packets only if they belong to a connection initiated inside	Much stronger than static filtering; still fast	Cannot inspect application content; state table is a DoS target
Application-level gateway / proxy	7	A proxy terminates the connection and re-originates it, understanding the specific protocol (HTTP, FTP) and inspecting content	Deep inspection , logging, content filtering, hides internal hosts	Slow ; a separate proxy is needed per application
Circuit-level gateway	5	Validates the TCP handshake, then relays bytes without inspecting them (SOCKS)	Low overhead, hides internal addresses	No content inspection
NGFW / UTM	3–7	Combines stateful inspection with IPS, deep packet inspection, application awareness, TLS inspection	Comprehensive	Costly, complex, can be a bottleneck

DMZ (Demilitarised Zone) — a subnet between two firewalls (or on a third interface of one) holding **public-facing servers** (web, mail, DNS). If a DMZ server is compromised, the attacker is still separated from the internal LAN by another firewall. Draw this: Internet → outer firewall → DMZ (web/mail servers) → inner firewall → internal LAN.

What a firewall cannot do — worth two marks: it cannot stop attacks that **bypass** it (rogue modems, USB drives, personal hotspots), it cannot stop **insider** attacks, it cannot detect malware in **encrypted** traffic without TLS interception, and it cannot patch a vulnerable application. Complement with **IDS/IPS**, **antivirus** and patching.

IDS vs IPS: an **Intrusion Detection System** monitors and **alerts** (passive, out of band); an **Intrusion Prevention System** sits **in line** and **blocks**. Detection methods: **signature-based** (matches known patterns; misses zero-days) and **anomaly/behaviour-based** (flags deviations from a baseline; more false positives).

VPN: a **Virtual Private Network** creates an **encrypted tunnel** across a public network, so a remote user or branch office behaves as if directly attached to the private LAN. Protocols: **IPSec** (**AH** for authentication/integrity, **ESP** for encryption; **transport mode** protects the payload, **tunnel mode** encapsulates the entire original packet in a new one), **SSL/TLS VPN**, PPTP, L2TP, WireGuard.

Tunnelling, fragmentation and internetworking

These three are named explicitly in your Forensic syllabus, so they need clean definitions.

Tunnelling is **encapsulating one protocol's packet inside another protocol's packet** so that it can traverse a network that does not support it. The classic example: two IPv6 islands separated by an IPv4-only Internet — each IPv6 packet is placed inside an IPv4 packet at the entry router, carried across as ordinary IPv4 payload, and unwrapped at the exit router. The intermediate network never knows IPv6 exists. Applications: **6to4 / Teredo** (IPv6 over IPv4), **VPNs** (IPSec tunnel mode, L2TP, PPTP, GRE), **MPLS**, connecting two remote LANs of the same type over a WAN of a different type. Analogy that works well in an answer: putting a car on a train to cross a mountain — the car doesn't drive through, it is *carried*.

Fragmentation happens because different networks have different **MTUs** (Maximum Transmission Unit — the largest payload a link can carry; **Ethernet's MTU is 1500 bytes**). When a router must forward a packet onto a link with a smaller MTU, it **breaks the packet into fragments**, each with its own IP header.

- IPv4 header fields used: **Identification** (all fragments of one packet share it), **Flags** — **DF** (Don't Fragment) and **MF** (More Fragments; set on all but the last), and **Fragment Offset** (position of this fragment's data, **measured in 8-byte units**).
- **Reassembly is done only at the final destination**, never at intermediate routers, because fragments may take different routes.
- **Costs**: more headers (overhead), more processing, and **loss of any one fragment forces retransmission of the whole original packet**.
- **Transparent fragmentation** — the fragments are reassembled at the exit gateway of the small-MTU network, so the next network sees the original packet.
Non-transparent fragmentation — fragments travel independently all the way to the destination host, which reassembles. **IP uses non-transparent fragmentation**.
- **IPv6 abolishes router fragmentation**: only the **source** may fragment, and it uses **Path MTU Discovery** (send with DF set; a router that cannot forward returns ICMP "fragmentation needed", revealing the path MTU).
- **Forensic relevance**: overlapping and tiny fragments are a classic technique for **evading IDS/firewalls** (teardrop attack, fragment overlap).

Worked fragmentation example. A 4000-byte IP packet (20-byte header + 3980 bytes of data) must cross a link with **MTU 1500**.

- Each fragment may carry at most $1500 - 20 = 1480$ bytes of data, and the data length of every fragment except the last must be a **multiple of 8**. 1480 is a multiple of 8 ✓

- Fragment 1: data bytes 0–1479, **offset = 0**, MF = 1, total length 1500
- Fragment 2: data bytes 1480–2959, **offset = 1480/8 = 185**, MF = 1, total length 1500
- Fragment 3: data bytes 2960–3979 (1020 bytes), **offset = 2960/8 = 370**, **MF = 0**, total length 1040
- Total: **3 fragments**; overhead rose from 20 bytes of header to 60.

Likely exam questions

- **[5]** Differentiate between active and passive attacks with examples.
- **[5]** Differentiate between symmetric and asymmetric cryptography.
- **[5]** What is a digital signature? How does it provide non-repudiation?
- **[5]** What is a firewall? Explain any two types.
- **[5]** Differentiate between a virus, a worm and a Trojan horse.
- **[5]** What is tunnelling? Give one application.
- **[5]** Explain fragmentation. What is an MTU?
- **[10]** Explain public key cryptography. How is it used for confidentiality and for authentication? Describe the working of RSA and of digital signatures with a diagram.
- **[10]** Explain firewalls: types, working, placement, the DMZ, and their limitations.
- **[15]** Discuss network security in detail: security goals, active and passive attacks, symmetric and asymmetric cryptography with a comparison, hash functions, digital signatures and certificates, firewalls and VPNs.

MCQ traps

Trap	Correct answer
"For confidentiality, encrypt with whose key?"	The receiver's public key.
"For a digital signature, encrypt with whose key?"	The sender's private key.
"Which provides non-repudiation?"	Asymmetric / digital signature , never symmetric.
"Keys needed for n users, symmetric?"	$n(n-1)/2$. Asymmetric = 2n .
"DES key size?"	56 bits (64 with parity). AES = 128/192/256.
"MD5 / SHA-1 digest size?"	128 / 160 bits.
"RSA's hard problem?"	Factoring large numbers . Diffie–Hellman → discrete log .
"Diffie–Hellman encrypts data?"	No — it is key exchange only.
"Does signing hide the message?"	No .
"Which firewall type inspects content?"	Application-level gateway (proxy) , layer 7.
"Ethernet MTU?"	1500 bytes .
"Fragment offset is measured in?"	8-byte units .
"Who reassembles IP fragments?"	The final destination only.
"Virus vs worm?"	A worm needs no host file and self-propagates .